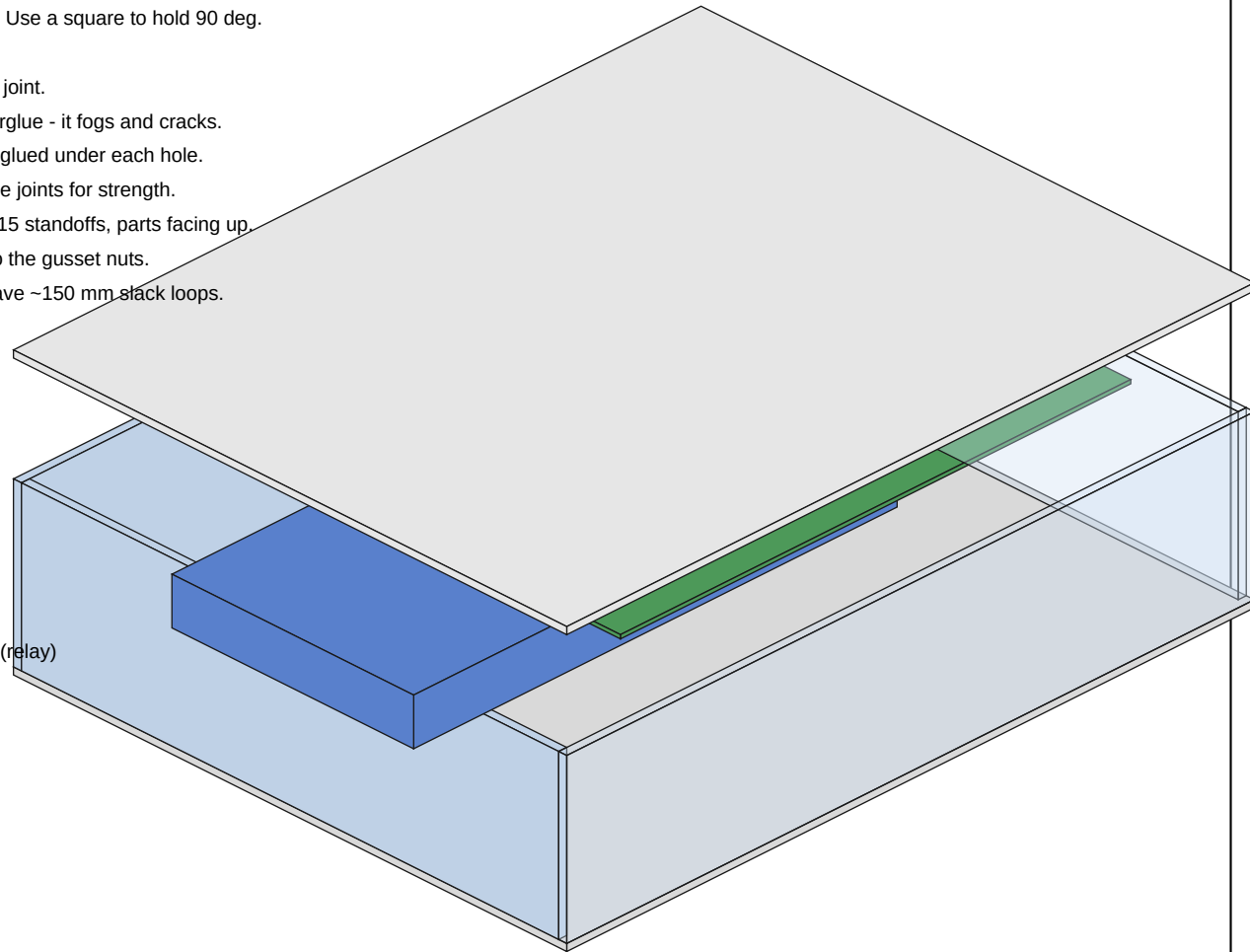


Assembly steps

1. Cut all parts (RED lines cut through, BLUE lines engrave or just mark). Peel the protective film only where you glue.
2. Check each part against its drawing sheet. Sheet thickness must be 3 mm - thicker sheet changes every size.
3. Fit cable glands, the relay-board standoffs (base) and the 5 PCB standoffs (lid) BEFORE gluing - easier flat.
4. Glue FRONT and BACK walls onto the base (walls stand ON the base, flush with its edges). Use a square to hold 90 deg.
5. Glue LEFT and RIGHT sides between the front and back walls, also standing on the base.
Acrylic: solvent cement (Weld-On 4 / dichloromethane) with a needle bottle, capillary into the joint.
PVC / ABS sheet: PVC cement. Polycarbonate or unknown plastic: 2-part epoxy. Avoid superglue - it fogs and cracks.
6. Glue 2 stacked gussets into each top inside corner, top face flush with the wall tops; M3 nut glued under each hole.
7. Let it cure 24 h before mounting boards. Run a thin fillet of glue/silicone along the inside base joints for strength.
8. Relay board on 4x M3 x 6 standoffs on the base. Main PCB hangs under the lid on 5x M3 x 15 standoffs, parts facing up.
9. Plug the 20-way ribbon (PCB J20 -> relay inputs) before closing. Lid: 4x M3 x 10 screws into the gusset nuts.
10. Field wiring: flip the lid over on the bench (PCB facing you), land wires on the terminals, leave ~150 mm slack loops.

Parts list

- 1 x BASE 256 x 206 mm
- 1 x LID 256 x 206 mm
- 1 x FRONT WALL 256 x 70 mm
- 1 x BACK WALL 256 x 70 mm
- 1 x LEFT SIDE 200 x 70 mm
- 1 x RIGHT SIDE 200 x 70 mm
- 8 x GUSSET 20 x 20 mm
- 5 x PG9 cable gland (hole 15.2) 6 x PG7 cable gland (hole 12.5)
- 5 x M3 x 15 standoff + 10 x M3 x 6 screw (PCB to lid) 4 x M3 x 6 standoff + 8 x M3 x 6 screw (relay)
- 4 x M3 x 10 screw + 4 x M3 nut (lid) 20-way IDC ribbon ~150 mm solvent cement / epoxy



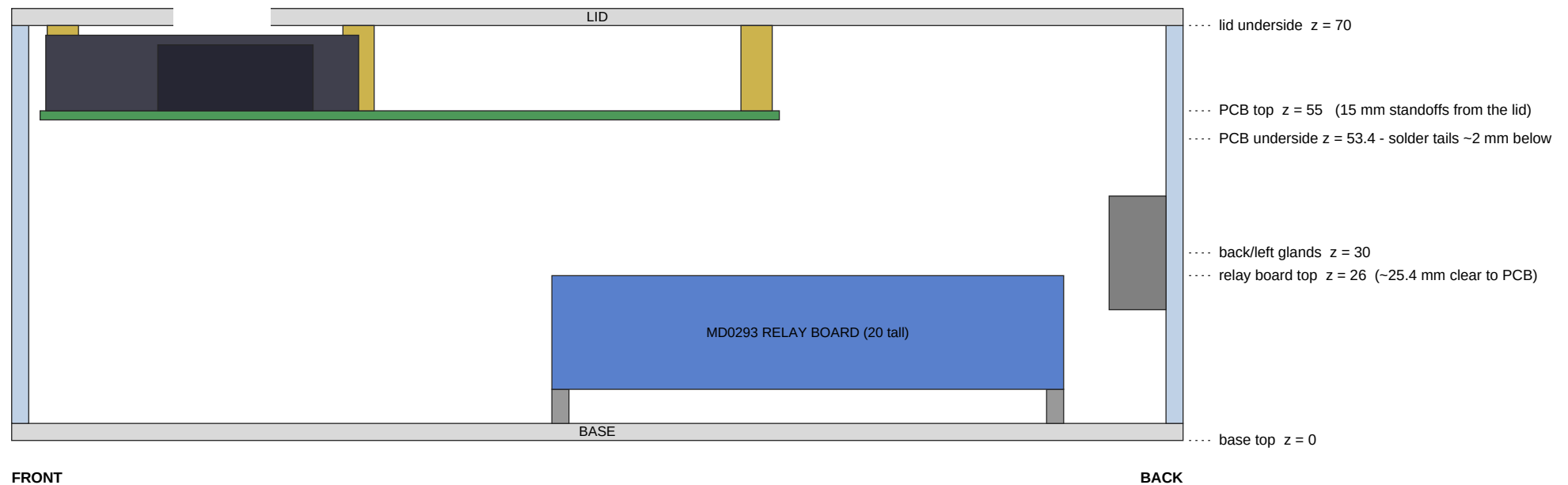
ASSEMBLY - outer box 256 x 206 x 76 mm

Plant Automation room controller enclosure | 3 mm plastic sheet, glued butt joints | all dimensions mm, origin = lower-left of the view

Dark blocks under the lid: ESP32 on sockets (13.3 tall, front) and OLED (11.6 tall, under the window). Gold: M3 standoffs.

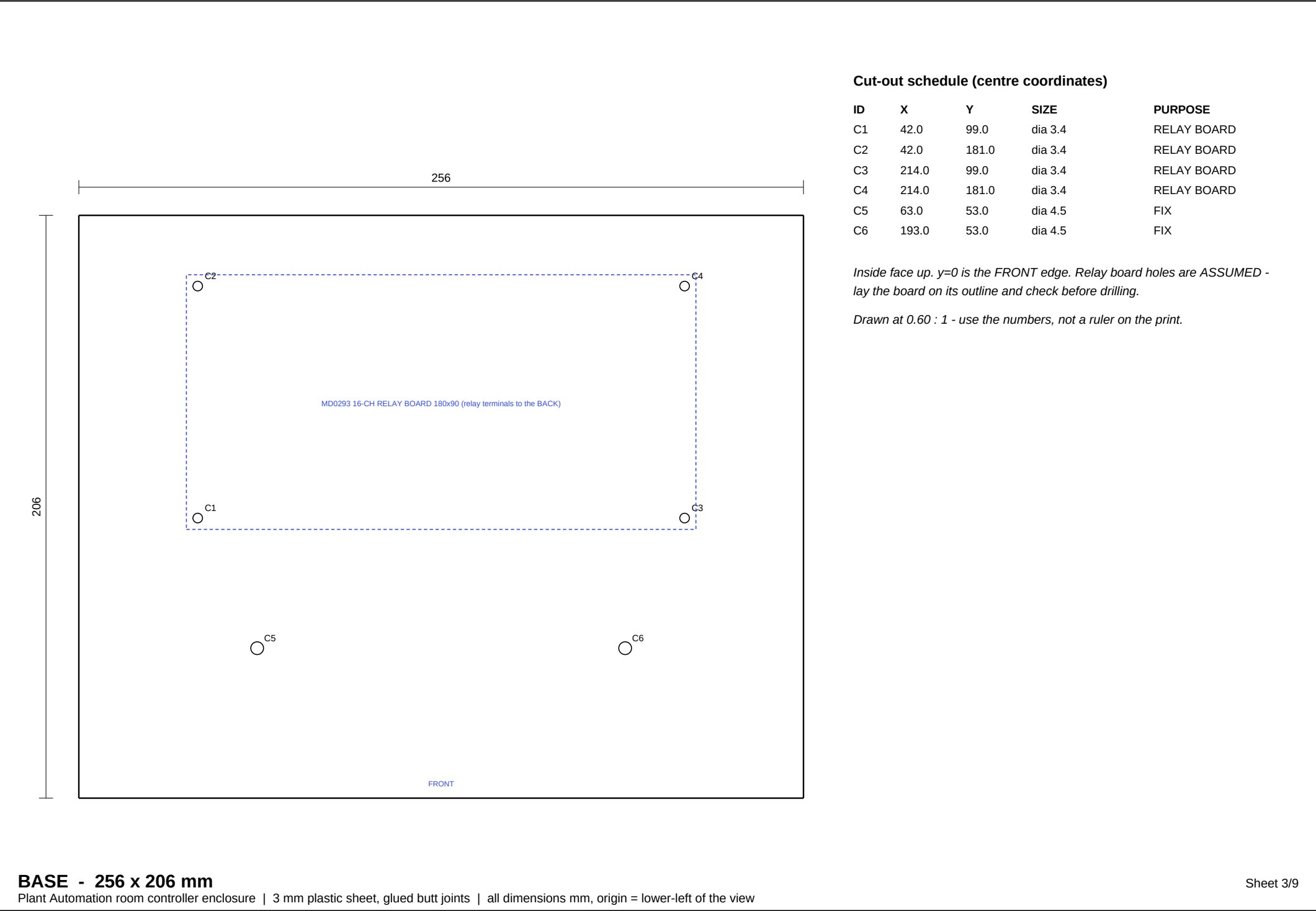
OLED glass ~3.4 mm below the lid window | ESP32 top clears the lid by ~1.7 mm | USB notch in the front wall at $z = 66.6$

Tallest parts on the PCB (verify on your modules): ESP32 on 8.5 mm sockets 13.3 | 74HC595 module on sockets ~14.6 | fuse holder ~12 | terminals ~10



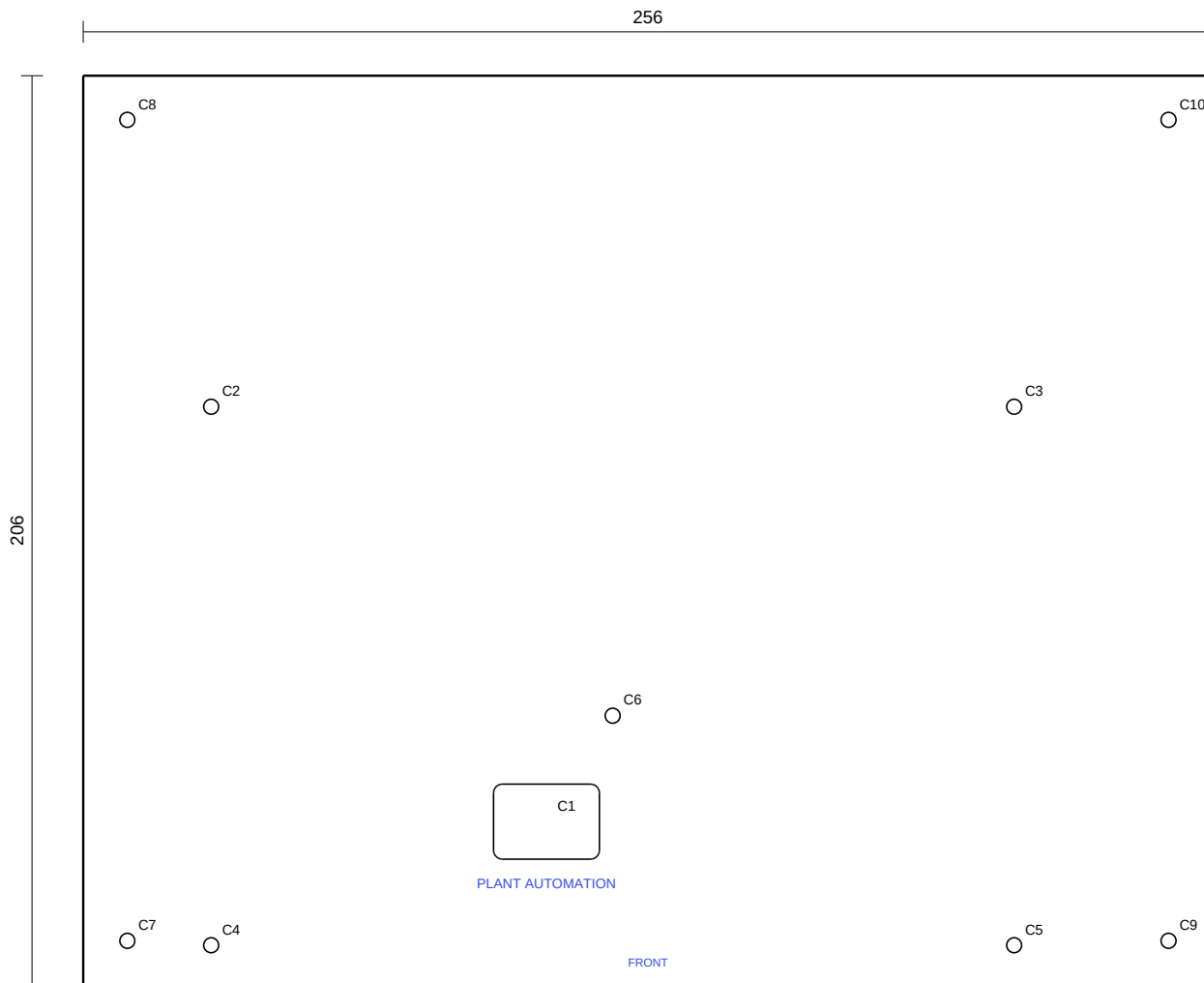
SECTION A-A - heights and clearances (side view from the left, front to the LEFT)

Plant Automation room controller enclosure | 3 mm plastic sheet, glued butt joints | all dimensions mm, origin = lower-left of the view



BASE - 256 x 206 mm

Plant Automation room controller enclosure | 3 mm plastic sheet, glued butt joints | all dimensions mm, origin = lower-left of the view



Cut-out schedule (centre coordinates)

ID	X	Y	SIZE	PURPOSE
C1	105.0	37.0	24 x 17 R2	OLED WINDOW
C2	29.0	131.0	día 3.4	PCB STANDOFF
C3	211.0	131.0	día 3.4	PCB STANDOFF
C4	29.0	9.0	día 3.4	PCB STANDOFF
C5	211.0	9.0	día 3.4	PCB STANDOFF
C6	120.0	61.0	día 3.4	PCB STANDOFF
C7	10.0	10.0	día 3.4	LID SCREW
C8	10.0	196.0	día 3.4	LID SCREW
C9	246.0	10.0	día 3.4	LID SCREW
C10	246.0	196.0	día 3.4	LID SCREW

Outside face up. y=0 is the FRONT edge. The main PCB hangs under it on 5x M3 x 15 standoffs.

Drawn at 0.60 : 1 - use the numbers, not a ruler on the print.

LID - 256 x 206 mm

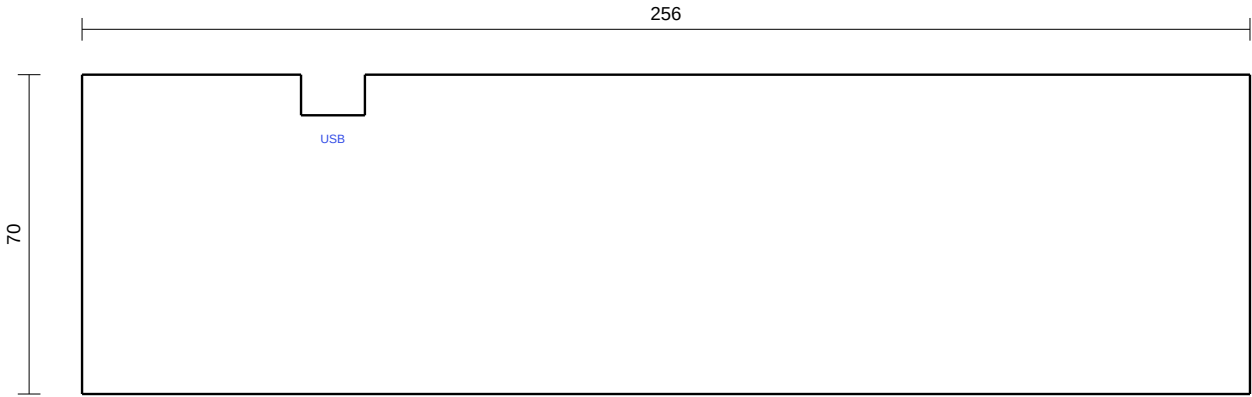
Plant Automation room controller enclosure | 3 mm plastic sheet, glued butt joints | all dimensions mm, origin = lower-left of the view

Cut-out schedule (centre coordinates)

ID	X	Y	SIZE	PURPOSE
N1	55.0	65.5	14 x 8.9 notch	USB (open at top)

Viewed from OUTSIDE. y=0 is the bottom edge (sits on the base). The USB notch opens at the top edge - the lid closes it.

Drawn at 0.60 : 1 - use the numbers, not a ruler on the print.



FRONT WALL - 256 x 70 mm

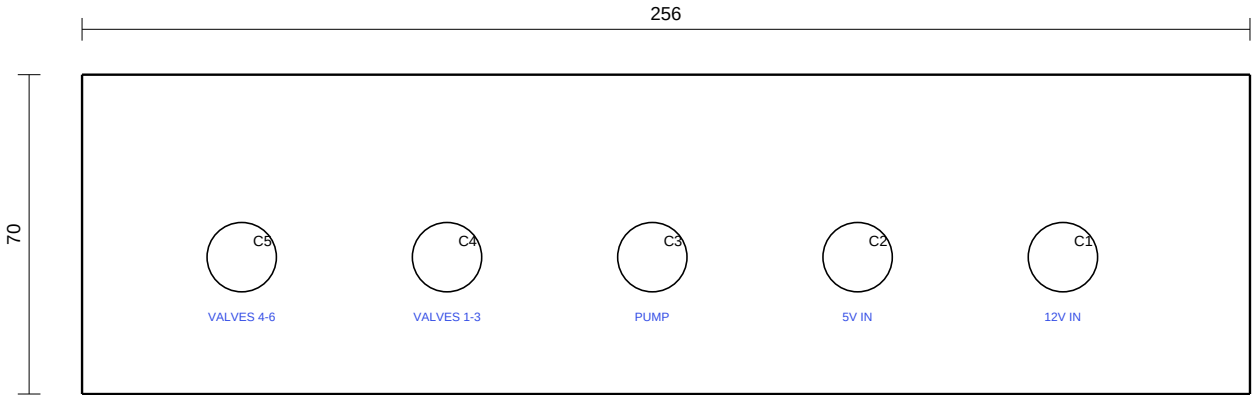
Plant Automation room controller enclosure | 3 mm plastic sheet, glued butt joints | all dimensions mm, origin = lower-left of the view

Cut-out schedule (centre coordinates)

ID	X	Y	SIZE	PURPOSE
C1	215.0	30.0	dia 15.2	PG9
C2	170.0	30.0	dia 15.2	PG9
C3	125.0	30.0	dia 15.2	PG9
C4	80.0	30.0	dia 15.2	PG9
C5	35.0	30.0	dia 15.2	PG9

Viewed from OUTSIDE. y=0 is the bottom edge. Valve glands take 3 cables each - use multi-hole gland inserts or a 6-core cable.

Drawn at 0.60 : 1 - use the numbers, not a ruler on the print.



BACK WALL - 256 x 70 mm

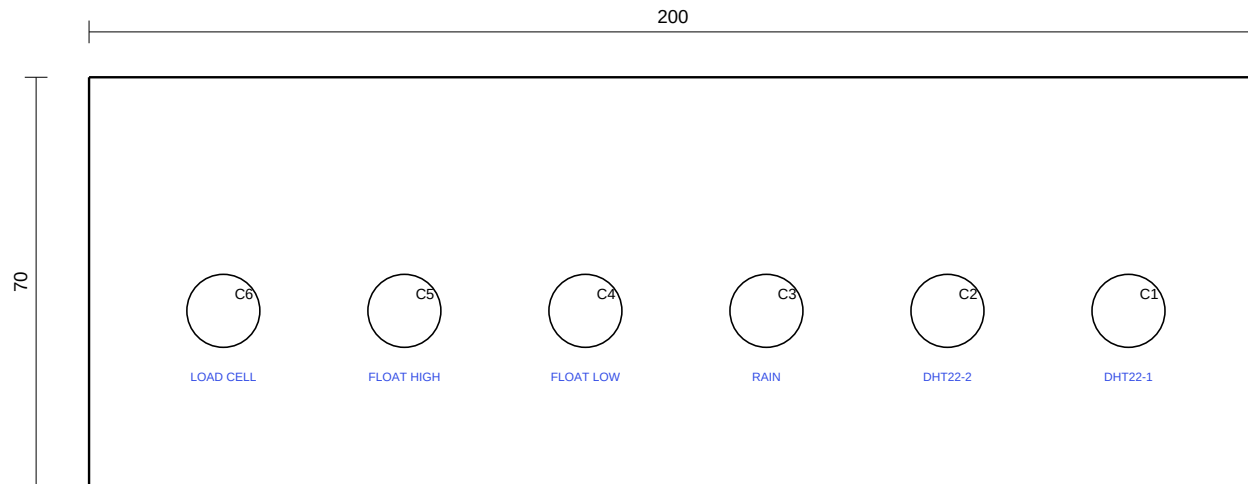
Plant Automation room controller enclosure | 3 mm plastic sheet, glued butt joints | all dimensions mm, origin = lower-left of the view

Cut-out schedule (centre coordinates)

ID	X	Y	SIZE	PURPOSE
C1	178.0	30.0	dia 12.5	PG7
C2	147.0	30.0	dia 12.5	PG7
C3	116.0	30.0	dia 12.5	PG7
C4	85.0	30.0	dia 12.5	PG7
C5	54.0	30.0	dia 12.5	PG7
C6	23.0	30.0	dia 12.5	PG7

Viewed from OUTSIDE. Fits between front and back walls; right-hand end is the FRONT.

Drawn at 0.77 : 1 - use the numbers, not a ruler on the print.



LEFT SIDE - 200 x 70 mm

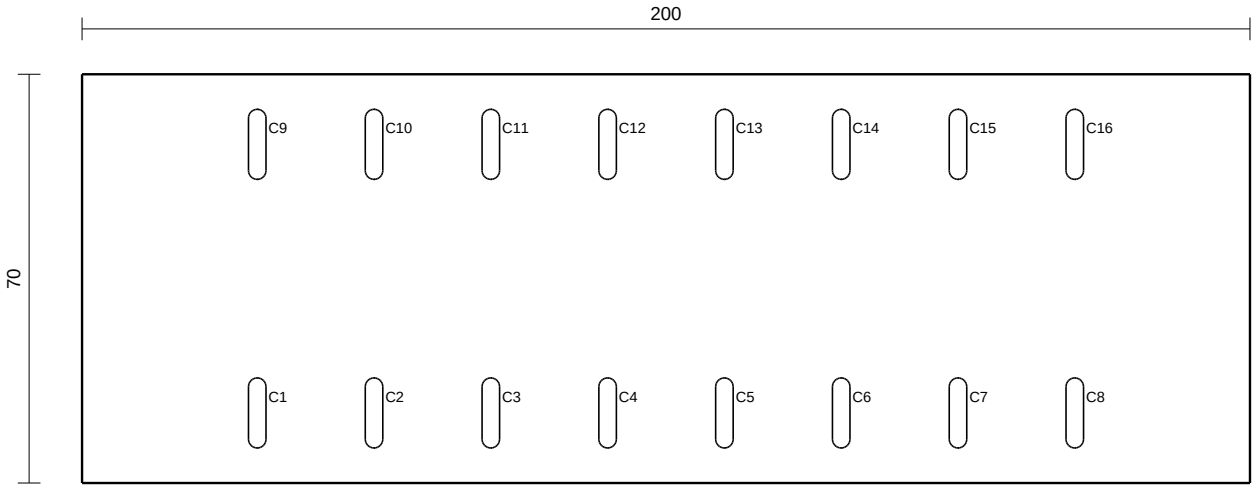
Plant Automation room controller enclosure | 3 mm plastic sheet, glued butt joints | all dimensions mm, origin = lower-left of the view

Cut-out schedule (centre coordinates)

ID	X	Y	SIZE	PURPOSE
C1	30.0	12.0	3 x 12 R1.5	VENT
C2	50.0	12.0	3 x 12 R1.5	VENT
C3	70.0	12.0	3 x 12 R1.5	VENT
C4	90.0	12.0	3 x 12 R1.5	VENT
C5	110.0	12.0	3 x 12 R1.5	VENT
C6	130.0	12.0	3 x 12 R1.5	VENT
C7	150.0	12.0	3 x 12 R1.5	VENT
C8	170.0	12.0	3 x 12 R1.5	VENT
C9	30.0	58.0	3 x 12 R1.5	VENT
C10	50.0	58.0	3 x 12 R1.5	VENT
C11	70.0	58.0	3 x 12 R1.5	VENT
C12	90.0	58.0	3 x 12 R1.5	VENT
C13	110.0	58.0	3 x 12 R1.5	VENT
C14	130.0	58.0	3 x 12 R1.5	VENT
C15	150.0	58.0	3 x 12 R1.5	VENT
C16	170.0	58.0	3 x 12 R1.5	VENT

Viewed from OUTSIDE. Vent slots: low row draws air in, high row lets heat out.

Drawn at 0.77 : 1 - use the numbers, not a ruler on the print.



RIGHT SIDE - 200 x 70 mm

Plant Automation room controller enclosure | 3 mm plastic sheet, glued butt joints | all dimensions mm, origin = lower-left of the view

Cut-out schedule (centre coordinates)

ID	X	Y	SIZE	PURPOSE
C1	7.0	7.0	dia 3.4	M3

Glue 2 stacked per top inside corner, flush with the wall tops; M3 nut under the lower one.

Drawn at 1.00 : 1 - use the numbers, not a ruler on the print.

